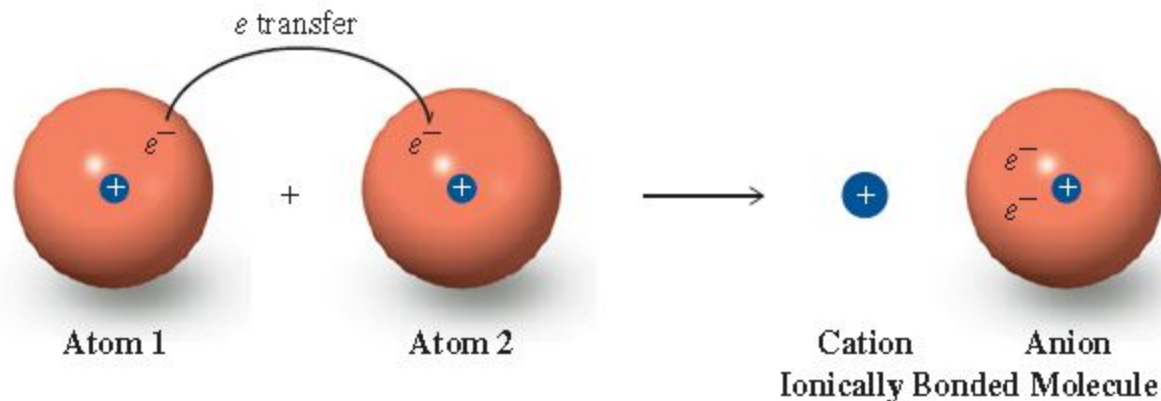
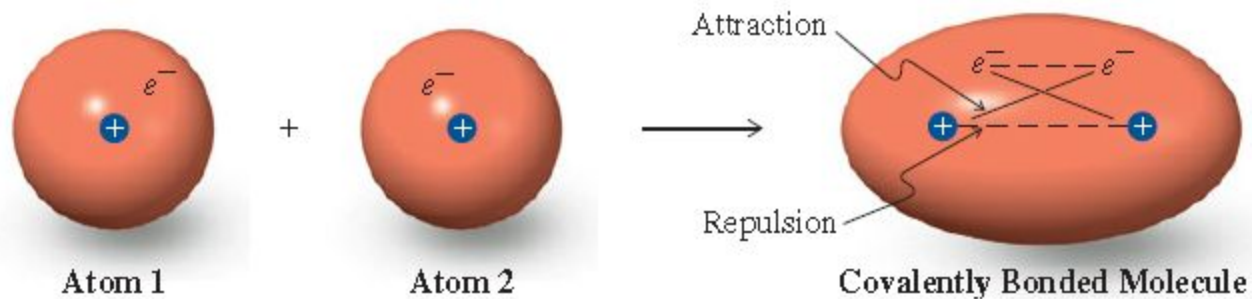


Chemical Bonding I

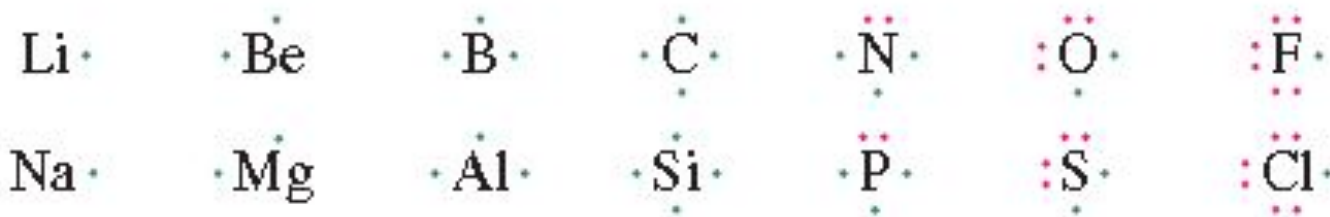
Chapter 9



Lewis Structures or, Lewis Dot Symbol

- **Lewis dot symbol** consists of the symbol of an element and one dot for each valence electron in an atom of the element
- In **Lewis structures**, an atom's valence electrons are **represented by dots**.
- **Valence electrons:** Outer shell electrons
 - Hydrogen has one dot ($1s$)
 - Carbon has four dots ($2s^2 2p^2$)
 - Oxygen has six dots ($2s^2 2p^4$)

Valence Electrons as Electron Dots



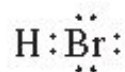
Lewis Structures

1 1A	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
·H·												·B·	·C·	·N·	·O·	·F·	·Ne·
·Li·	·Be·											·Al·	·Si·	·P·	·S·	·Cl·	·Ar·
·Na·	·Mg·	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	·Ga·	·Ge·	·As·	·Se·	·Br·	·Kr·
·K·	·Ca·											·In·	·Sn·	·Sb·	·Te·	·I·	·Xe·
·Rb·	·Sr·											·Tl·	·Pb·	·Bi·	·Po·	·At·	·Rn·
·Cs·	·Ba·																
·Fr·	·Ra·																

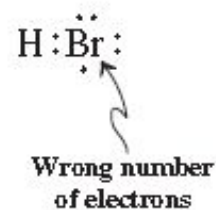
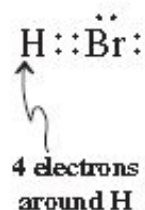
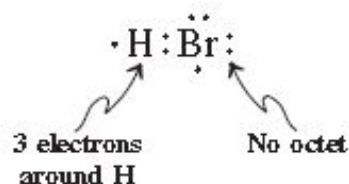
Figure 9.1

Lewis dot symbols for the representative elements and the noble gases. The number of unpaired dots corresponds to the number of bonds an atom of the element can form in a compound.

Correct Lewis Structure



Incorrect Lewis Structures

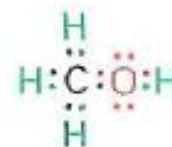
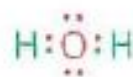
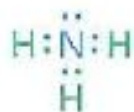
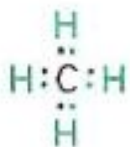


Kekulé structures or line-bond structures

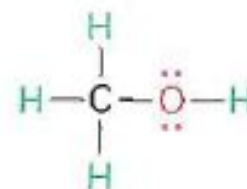
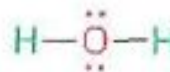
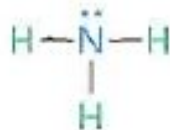
In **Kekule structures**, the two electrons of a covalent bond are **indicated by a line** between the atoms.

A stable molecule results when a **noble-gas configuration** is achieved for all atoms- an **octet (eight dots)** for most elements or **two for hydrogen**.

Electron-dot structures
(Lewis structures)



Line-bond structures
(Kekulé structures)



Methane
(CH₄)

Ammonia
(NH₃)

Water
(H₂O)

Methanol
(CH₃OH)

Writing Lewis Structures

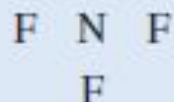
1. Write the skeletal structure of the compound, using chemical symbols and placing bonded atoms next to one another. For simple compounds, this task is fairly easy. For more complex compounds, we must either be given the information or make an intelligent guess about it. In general, the least electronegative atom occupies the central position. Hydrogen and fluorine usually occupy the terminal (end) positions in the Lewis structure.
2. Count the total number of valence electrons present, referring, if necessary, to Figure 9.1. For polyatomic anions, add the number of negative charges to that total. (For example, for the CO_3^{2-} ion we add two electrons because the $2-$ charge indicates that there are two more electrons than are provided by the atoms.) For polyatomic cations, we subtract the number of positive charges from this total. (Thus, for NH_4^+ we subtract one electron because the $1+$ charge indicates a loss of one electron from the group of atoms.)
3. Draw a single covalent bond between the central atom and each of the surrounding atoms. Complete the octets of the atoms bonded to the central atom. (Remember that the valence shell of a hydrogen atom is complete with only two electrons.) Electrons belonging to the central or surrounding atoms must be shown as lone pairs if they are not involved in bonding. The total number of electrons to be used is that determined in step 2.
4. After completing steps 1–3, if the central atom has fewer than eight electrons, try adding double or triple bonds between the surrounding atoms and the central atom, using lone pairs from the surrounding atoms to complete the octet of the central atom.

EXAMPLE 9.3

Write the Lewis structure for nitrogen trifluoride (NF_3) in which all three F atoms are bonded to the N atom.

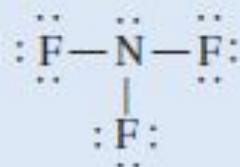
Solution We follow the preceding procedure for writing Lewis structures.

Step 1: The N atom is less electronegative than F, so the skeletal structure of NF_3 is



Step 2: The outer-shell electron configurations of N and F are $2s^22p^3$ and $2s^22p^5$, respectively. Thus, there are $5 + (3 \times 7)$, or 26, valence electrons to account for in NF_3 .

Step 3: We draw a single covalent bond between N and each F, and complete the octets for the F atoms. We place the remaining two electrons on N:



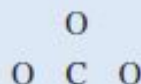
Because this structure satisfies the octet rule for all the atoms, step 4 is not required.

EXAMPLE 9.5

Write the Lewis structure for the carbonate ion (CO_3^{2-}).

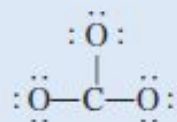
Solution We follow the preceding procedure for writing Lewis structures and note that this is an anion with two negative charges.

Step 1: We can deduce the skeletal structure of the carbonate ion by recognizing that C is less electronegative than O. Therefore, it is most likely to occupy a central position as follows:



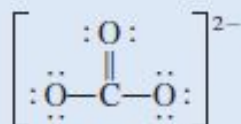
Step 2: The outer-shell electron configurations of C and O are $2s^2 2p^2$ and $2s^2 2p^4$, respectively, and the ion itself has two negative charges. Thus, the total number of electrons is $4 + (3 \times 6) + 2$, or 24.

Step 3: We draw a single covalent bond between C and each O and comply with the octet rule for the O atoms:



This structure shows all 24 electrons.

Step 4: Although the octet rule is satisfied for the O atoms, it is not for the C atom. Therefore we move a lone pair from one of the O atoms to form another bond with C. Now the octet rule is also satisfied for the C atom:



Bonding

Octet rule: The tendency among atoms of Group 1A-7A elements to react in ways **that achieve an outer shell of eight valence electrons**

- An atom that gains electrons becomes an anion.
- An atom that loses electrons becomes a cation.

Ionic bond: A chemical bond resulting from **the electrostatic attraction** of an anion and a cation.

Covalent bond: A chemical bond resulting from **two atoms sharing one or more pairs of electrons**.

Electronegativity: A measure of the force of an atom's attraction for the electrons it shares in a chemical bond with another atom.

- Electronegativity increases from left to right within a period.
- Electronegativity Increases from bottom to top in a group.

Electronegativity increases									
Li	Be				B	C	N	O	F
Na	Mg				Al	Si	P	S	Cl
K	Ca								Br
									I

Electronegativity increases

Electronegativity Values for Some Atoms (Pauling Scale)

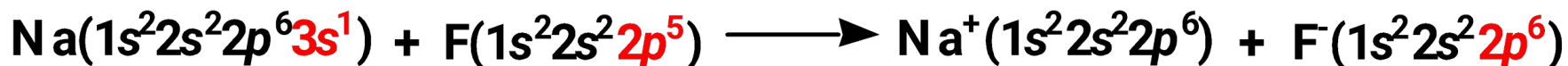
TABLE 1.5 Electronegativity Values for Some Atoms (Pauling Scale)

Electronegativity																
1A	2A	H 2.1										3A	4A	5A	6A	7A
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2	3B	4B	5B	6B	7B	8B			1B	2B	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2
Legend																
<1.0	1.0 – 1.4	1.5 – 1.9	2.0 – 2.4		2.5 – 2.9	3.0 – 4.0										

Ionic Bond

Ionic Bond:

An ionic bond forms by the transfer of electrons from the valence shell of an atom of lower electronegativity to the valence shell of an atom of higher electronegativity.



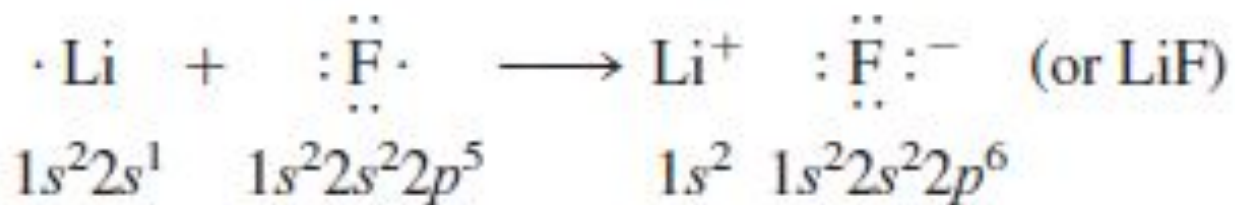
Consider the reaction between lithium and fluorine to form lithium fluoride

The electronic configuration of lithium is $1s^2 2s^1$

The electronic configuration of fluorine is $1s^2 2s^2 2p^5$

When lithium and fluorine atoms come in contact with each other, the outer $2s^1$ valence electron of lithium is transferred to the fluorine atom.

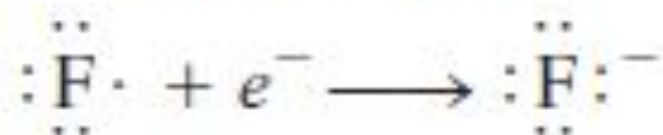
Using Lewis dot symbols, we represent the reaction as follows:



For convenience, imagine that this reaction occurs in separate steps—first the ionization of Li:



and then the acceptance of an electron by F:



Next, imagine the two separate ions joining to form a LiF unit:

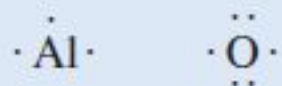


EXAMPLE 9.1

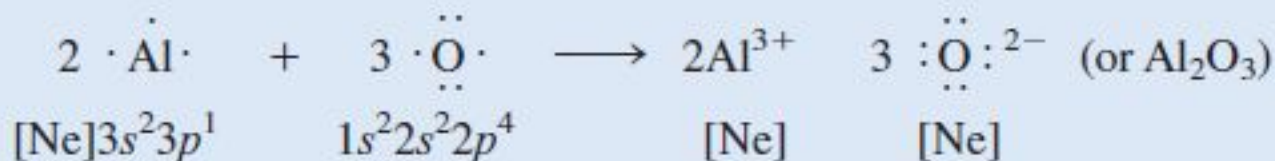
Use Lewis dot symbols to show the formation of aluminum oxide (Al_2O_3).

Strategy We use electroneutrality as our guide in writing formulas for ionic compounds, that is, the total positive charges on the cations must be equal to the total negative charges on the anions.

Solution According to Figure 9.1, the Lewis dot symbols of Al and O are



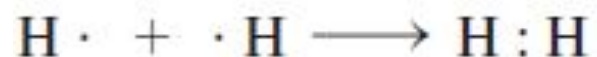
Because aluminum tends to form the cation (Al^{3+}) and oxygen the anion (O^{2-}) in ionic compounds, the transfer of electrons is from Al to O. There are three valence electrons in each Al atom; each O atom needs two electrons to form the O^{2-} ion, which is isoelectronic with neon. Thus, the simplest neutralizing ratio of Al^{3+} to O^{2-} is 2:3; two Al^{3+} ions have a total charge of +6, and three O^{2-} ions have a total charge of -6. So the empirical formula of aluminum oxide is Al_2O_3 , and the reaction is



Covalent Bond:

A covalent bond forms when electron pairs are shared between two atoms.

Formation of a covalent bond in H_2

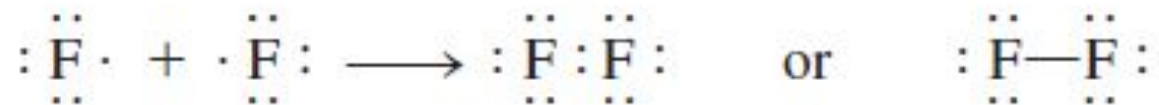


Consider the formation of fluorine molecule, F_2

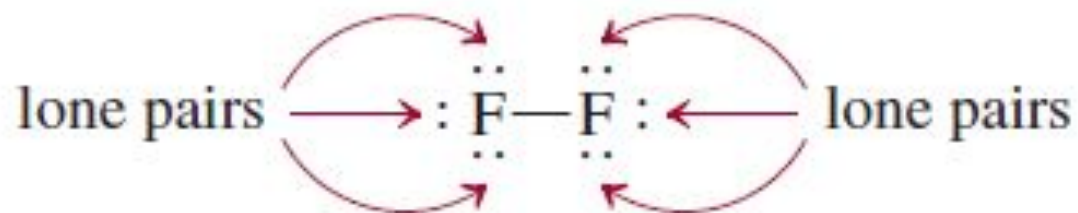
The electronic configuration of F is $1s^2 2s^2 2p^5$

Each F atom has 7 valence electrons (the 2s and 2p electrons)

Formation of F_2 is as follows:



Lone pairs: Pairs of valence electrons that are not involved in covalent bond Formation.



Formal Charge

Formal charge: The **charge on an atom in a molecule** or polyatomic ion is called its formal charge

$$\text{Formal charge} = \left(\begin{array}{c} \text{number of outer-shell} \\ \text{electrons on the} \\ \text{free, neutral atom} \end{array} \right) - \left(\begin{array}{c} \text{number of unshared} \\ \text{electrons on the atom} \\ \text{in the molecule} \end{array} \right) - \frac{1}{2} \left(\begin{array}{c} \text{number of bonding} \\ \text{electrons surrounding the} \\ \text{atom in the molecule} \end{array} \right)$$

or simply

$$\begin{aligned} \text{Formal charge} = & \text{number of valence electrons} - \text{number of lone pair electrons} \\ & - \frac{1}{2} \text{ number of bonding electrons} \end{aligned}$$

To assign the number of electrons on an atom in a Lewis structure, we proceed as follows:

- All the atom's nonbonding electrons are assigned to the atom.
- We break the bond(s) between the atom and other atom(s) and assign half of the bonding electrons to the atom.

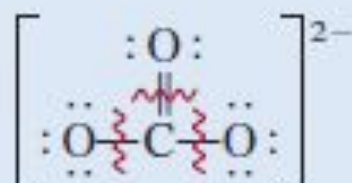
When you write formal charges, these rules are helpful:

1. For molecules, the sum of the formal charges must add up to zero because molecules are electrically neutral species. (This rule applies, for example, to the O_3 molecule.)
2. For cations, the sum of the formal charges must equal the positive charge.
3. For anions, the sum of the formal charges must equal the negative charge.

EXAMPLE 9.6

Write formal charges for the carbonate ion.

Solution The Lewis structure for the carbonate ion was developed in Example 9.5:



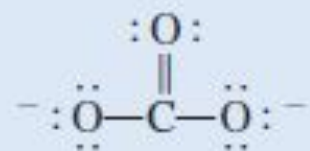
The formal charges on the atoms can be calculated using the given procedure.

The C atom: The C atom has four valence electrons and there are no nonbonding electrons on the atom in the Lewis structure. The breaking of the double bond and two single bonds results in the transfer of four electrons to the C atom. Therefore, the formal charge is $4 - 4 = 0$.

The O atom in C=O: The O atom has six valence electrons and there are four nonbonding electrons on the atom. The breaking of the double bond results in the transfer of two electrons to the O atom. Here the formal charge is $6 - 4 - 2 = 0$.

The O atom in C—O: This atom has six nonbonding electrons and the breaking of the single bond transfers another electron to it. Therefore, the formal charge is $6 - 6 - 1 = -1$.

Thus, the Lewis structure for CO_3^{2-} with formal charges is



Check Note that the sum of the formal charges is -2 , the same as the charge on the carbonate ion.